JUTHELINK · PHASE 1

Set up your MikroTik hotspot and Tenda outdoor access point

A complete, first-time, click-by-click walkthrough. At the end, customers connect to “Starling Wi-Fi”, see your login page, type the username and password you sold them, and are disconnected automatically when their time is up.

Your equipment
Starlink · MikroTik hAP lite (RouterOS 6.48.6) · Tenda outdoor AP

Time needed
About 2–3 hours, with breaks. Do not rush.

Skill needed
None. Every click and every command is written here.

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Tip: This guide sets up the router to work **on its own** (phase 1): you create usernames and passwords, sell them for cash or Mobile Money, and the router does the rest. Later, the Starling website will be connected to it (phase 2). Nothing in this guide needs to be redone for that.

PART 0

Before you start

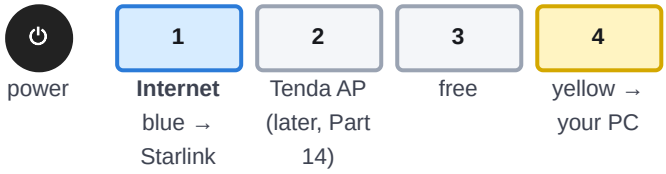
What you are building, what you need, and three rules that keep you safe.

What you are building

Starlink → blue cable → **MikroTik port 1** … MikroTik … **port 2** → **Tenda AP** → **customer phones**

The MikroTik sits in the middle. Internet comes in on **port 1**. Everything plugged into **ports 2, 3, 4** (and its own Wi-Fi) is the customer side: anyone there must log in on your page before they get internet.

Your cables (from your photo)



Looking at the back of the hAP lite. Port 1 is the one next to the power socket, marked “Internet”. Keep it like this for the whole guide.

What you need

- [ ] Starlink set up and working (the Starlink app shows **Online**).
- [ ] The MikroTik hAP lite, powered, blue cable in port 1, yellow cable in port 4 to your PC.
- [ ] Your PC with **Winbox** (you already have it and can connect by MAC address).
- [ ] A phone for testing.
- [ ] The kit `Juthelink-Router-Kit.zip` (unzip it on your PC): `starling-profiles.rsc`, `juthelink-sync.rsc`, `ticket-maker.html`, and the `hotspot` folder with `login.html`, `alogin.html`, `status.html`.
- [ ] The Tenda outdoor AP, its PoE injector and power adapter (only for Part 14).
- [ ] A notebook to write down passwords.

Three rules

Careful: Always connect Winbox using the MAC address (the letters/numbers like `18:FD:74:FF:17:EB`), not an IP address, until Part 9. The MAC connection keeps working even if you make a mistake with IP settings.

Careful: Type values exactly as written. Small letters stay small. No extra spaces. When this guide says `bridge-hotspot`, type exactly that.

Careful: If you get lost, you can always start again with Part 2 (start clean). Nothing you do here can damage the router.

Write your values here

What	Value
Router MAC address (Winbox)	
Router admin password (Part 3)	
Wi-Fi name (SSID)	Starling Wi-Fi
Hotspot admin test user / password (Part 8)	
Your WhatsApp / MoMo number (Part 12)	
Tenda admin password (Part 14)	

PART 1

Winbox basics

Five things you will use in every step.

1 Connecting

Open Winbox → **Neighbors** tab → **Refresh** → click the line that shows **hAP lite** → make sure the **MAC address** appears in “Connect To” → Login `admin`, Password: empty (or the password you set in Part 3) → **Connect**.

2 The menu on the left

Everything is in the left menu: **Interfaces**, **Wireless**, **Bridge**, **IP** (has a sub-menu: Addresses, DHCP Client, DNS, Hotspot, Firewall…), **System** (sub-menu: Password, Identity, Clock, Packages, Reset Configuration…), **Files**, **New Terminal**. In this guide, “**IP → DNS**” means: click IP, then DNS in the sub-menu.

3 Adding and editing

In most windows, the blue **+** (or **New**) button adds an item. Double-click a line to edit it. In an edit window, **Apply** saves and keeps it open, **OK** saves and closes.

4 The terminal (for copy-paste commands)

New Terminal opens a black window. You can paste a command there (right-click → Paste, or Ctrl+V) and press **Enter**. Every step in this guide shows both ways: the clicks, and the command. **Use one or the other for a step, not both.** If you are unsure, use the clicks.

5 Safe Mode (your undo button)

The **Safe Mode** button (top right in Winbox) protects you: while it is on, if Winbox loses the connection, the router automatically **undoes** your last changes. Turn it on before risky steps (this guide tells you when), and turn it off again when you are happy with the result so the changes are kept.

Tip: Your Winbox may look slightly different (newer Winbox 4 has a darker style and some buttons are named “New” instead of “+”). The menu names are the same.

PART 2

Start clean

Your router is almost empty already (it showed a `bridge1`). We erase everything so it is in a known state.

1. **Reset with no configuration**

   System → Reset Configuration → tick **No Default Configuration** → leave the other boxes unticked → **Reset Configuration** → confirm **Yes**.

   ```
   /system reset-configuration no-defaults=yes skip-backup=yes
   ```

   (In the terminal, type `y` and Enter when it asks “Dangerous! Reset anyway?”)

2. **Wait and reconnect**

   Winbox disconnects. Wait about 1 minute (the router reboots). Then Winbox → Neighbors → Refresh → click the MAC address → login `admin`, empty password → Connect. If it asks about the default configuration, click **Remove Configuration** (or OK).

✔ **You should see: Bridge** is empty, **IP → Addresses** is empty. That is correct.

✖ **If not:** The router does not appear in Neighbors: unplug the yellow cable from the PC and plug it back in, wait 20 seconds, Refresh again. Try another port (2 or 3) on the router.

PART 3

Password, name and clock

1. **Set the admin password**

   System → Password → Old Password: empty → New Password and Confirm: a strong password (write it in Part 0) → **Change**.

   ```
   /password old-password="" new-password="YourStrongPassword" confirm-new-password="YourStrongPassword"
   ```

2. **Give the router a name**

   System → Identity → `starling-hotspot` → OK.

   ```
   /system identity set name=starling-hotspot
   ```

3. **Time zone and automatic clock**

   System → Clock → Time Zone Name: `Africa/Douala` → OK. Then **System → SNTP Client** → tick **Enabled** → Primary NTP Server `162.159.200.1`, Secondary `162.159.200.123` → OK.

   ```
   /system clock set time-zone-name=Africa/Douala
   /system ntp client set enabled=yes primary-ntp=162.159.200.1 secondary-ntp=162.159.200.123
   ```

   The clock matters: customers' time is counted with it. It will show the right time once the internet works (Part 4).

Tip: From now on, log in with `admin` and your new password.

PART 4

Internet from Starlink (port 1)

The router asks the Starlink router for an address on port 1, exactly like a phone does.

---

1 Check Starlink

The Starlink app says **Online**, and the blue cable goes from a Starlink router Ethernet/LAN port to **port 1** of the MikroTik.

2 Add a DHCP client on port 1

IP → DHCP Client → **+** → Interface: `ether1` → **Use Peer DNS: unticked** → Add Default Route: **yes** → OK.

```
/ip dhcp-client add interface=ether1 use-peer-dns=no add-default-route=yes disabled=no
comment="Internet from Starlink"
```

✔ **You should see:** In the DHCP Client list, the Status becomes **bound** and an IP address appears (for example `192.168.1.23/24`, or `100.x.x.x` if Starlink is in bypass mode).

✖ **If not:** Status stays “searching…”: check the blue cable is in **port 1** and in the Starlink router; restart the Starlink router; wait 1 minute. If Starlink uses the Ethernet Adapter, check it is connected.

3 Set the DNS servers

IP → DNS → Servers: `1.1.1.1` and (click the small arrow to add a line) `8.8.8.8` → tick **Allow Remote Requests** → OK.

```
/ip dns set servers=1.1.1.1,8.8.8.8 allow-remote-requests=yes
```

4 Test the internet

New Terminal and type:

```
/ping google.com count=4
```

✔ **You should see:** Four lines with a time like `time=40ms` and “received=4”.

✖ **If not:** “could not resolve”: redo step 3. “timeout”: the DHCP client is not bound; redo step 2.

5 Check the clock

System → Clock: the date and time are now correct (Douala time). If not, wait 2 minutes and look again.

PART 5

Update the router

Always update before configuring the hotspot, so an update does not change your settings later. Your router has 6.48.6; we move to the latest 6.49 “long-term” version.

1 Download and install

System → Packages → Check For Updates → Channel: **long-term** → wait until it shows the Latest Version (6.49.something) → **Download&Install**.

```
/system package update set channel=long-term
/system package update check-for-updates
/system package update install
```

✔ **You should see:** The router downloads, then reboots by itself (Winbox disconnects). Wait 2–3 minutes, reconnect by MAC. **System → Packages** shows the new version.

2 Update the RouterBOARD firmware

System → RouterBOARD → if “Current Firmware” and “Upgrade Firmware” are different, click **Upgrade** → OK. Then **System → Reboot** → Yes. Reconnect after 1 minute.

```
/system routerboard upgrade
/system reboot
```

Careful: Do not unplug the power while it is updating. If the update fails or does not show 6.49, that is fine: continue with 6.48.6. Everything in this guide works on both.

Careful: Stay on version 6 (long-term). Do not choose the “stable/7” channel: the hAP lite has very little storage and this guide is written for version 6.

PART 6

Customer network (the bridge)

A “bridge” joins ports 2, 3, 4 and the Wi-Fi into one customer network. Port 1 (internet) must NOT be in it.

1 Create the bridge

Bridge → **Bridge** tab → **+** → Name: `bridge-hotspot` → OK.

```
/interface bridge add name=bridge-hotspot comment="Customers"
```

2 Put ports 2, 3, 4 and the Wi-Fi in it

Bridge → **Ports** tab → **+** → Interface `ether2`, Bridge `bridge-hotspot` → OK. Repeat for `ether3`, `ether4` and `wlan1`.

```
/interface bridge port add bridge=bridge-hotspot interface=ether2
/interface bridge port add bridge=bridge-hotspot interface=ether3
/interface bridge port add bridge=bridge-hotspot interface=ether4
/interface bridge port add bridge=bridge-hotspot interface=wlan1
```

✔ **You should see:** The Ports tab lists ether2, ether3, ether4, wlan1, all with bridge **bridge-hotspot**. **ether1 is not there.**

Tip: Winbox may disconnect for a second because your PC (port 4) just moved into the bridge. Reconnect by MAC. This is normal.

3 Give the customer network its address

IP → **Addresses** → **+** → Address `10.5.50.1/24` → Interface `bridge-hotspot` → OK.

```
/ip address add address=10.5.50.1/24 interface=bridge-hotspot comment="Customer gateway"
```

✔ **You should see:** IP → Addresses shows `10.5.50.1/24` on bridge-hotspot (and the Starlink address on ether1, marked D).

Tip: 10.5.50.1 is your router's address for customers. Customers will get addresses 10.5.50.10 to 10.5.50.250. The Tenda will later get 10.5.50.2.

PART 7

The router's own Wi-Fi (for testing now)

The hAP lite has a small indoor Wi-Fi. We turn it on so you can test everything before mounting the Tenda. You can keep it on afterwards as a second, indoor access point.

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1 **Make the Wi-Fi open (no password)**

The hotspot login page is the lock, so the Wi-Fi itself has no password (like airports and cafés).
Wireless → **Security Profiles** tab → double-click **default** → Mode: **none** → OK.

```
/interface wireless security-profiles set [find default=yes] mode=none
```

2 **Set the Wi-Fi name and turn it on**

Wireless → **WiFi Interfaces** tab → double-click **wlan1** → **Wireless** tab: Mode **ap bridge**, Band **2GHz-B/G/N**, Channel Width **20MHz**, Frequency **auto**, SSID `Starling Wi-Fi`, Security Profile **default** → OK. Then select wlan1 and click the blue tick ✓ **Enable**.

```
/interface wireless set wlan1 mode=ap-bridge band=2ghz-b/g/n channel-width=20mhz
frequency=auto ssid="Starling Wi-Fi" security-profile=default wireless-protocol=802.11
disabled=no
```

✔ **You should see:** On your phone, a Wi-Fi called **Starling Wi-Fi** appears, without a lock icon. Do not connect yet: there is no hotspot until Part 8, so the phone would not get an address.

PART 8

Turn on the hotspot (captive portal)

The Hotspot Setup wizard creates the login system, gives addresses to customers, and shares the internet. This is the same wizard as in the video.

1 Start the wizard

IP → Hotspot → **Servers** tab → **Hotspot Setup**. Answer each screen and click **Next**:

The wizard asks	You answer
HotSpot Interface	`bridge-hotspot`
Local Address of Network	`10.5.50.1/24  (Masquerade Network: ticked)`
Address Pool of Network	`10.5.50.10-10.5.50.250`
Select Certificate	`none`
IP Address of SMTP Server	`0.0.0.0`
DNS Servers	`1.1.1.1 and 8.8.8.8`
DNS Name	`login.starling`
Name of Local HotSpot User	`admin  + a password you choose (for your own tests; write it in Part 0)`

✔ **You should see:** "Setup has completed successfully". In the Servers tab there is one server on bridge-hotspot (Winbox names it **server1** or **hotspot1**; either is fine). **IP → DHCP Server** shows a server on bridge-hotspot. **IP → Firewall → NAT** shows a "masquerade hotspot network" rule.

Tip: The DNS Name `login.starling` is the address customers see at the top of the login page. It is not a real website; the router answers it itself.

2 Make sure FastTrack is not active

FastTrack is a speed shortcut that lets customers skip the login. You started clean, so there should be none, but check: **IP → Firewall → Filter Rules**. If any rule has Action **fasttrack connection**, select it and click the red **✗ Disable**.

```
/ip firewall filter disable [find action=fasttrack-connection]
```

3 First test with your phone

Phone → Wi-Fi → **Starling Wi-Fi** → connect. Open the browser and go to `http://neverssl.com`.

✔ **You should see:** MikroTik's grey login page appears (Part 12 replaces it with yours). Log in with `admin` and the hotspot password from step 1: you can browse.

✖ **If not:** **Phone says “Couldn't get IP address”** (and IP → DHCP Server → Leases stays empty): the Wi-Fi is not part of the customer bridge, or the wizard was run several times. Run this in New Terminal (safe to run more than once), then forget the network on the phone and join again:

```
/interface bridge port remove [find interface=wlan1]
/interface bridge port add bridge=bridge-hotspot interface=wlan1
/ip dhcp-server network remove [find]
/ip dhcp-server network add address=10.5.50.0/24 gateway=10.5.50.1 dns-server=10.5.50.1
/ip dhcp-server enable [find]
```

✖ **If not:** The phone says “no internet” and no page appears: type `http://login.starling` in the browser. Still nothing: check **IP → Hotspot → Servers**: hotspot1 must not be red (disabled). Check **IP → DHCP Server → Leases**: your phone must have a 10.5.50.x address.

PART 9

Let your own PC and phone skip the login

Your PC on port 4 is now on the customer network, so its browser would also need to log in. You can let your own devices through without a ticket.

1. **Find your PC's MAC address**

   IP → **DHCP Server** → **Leases** tab. Your PC is the line with your PC's name (Host Name). Note its **MAC Address**. (Also your phone's, from the test in Part 8.)

2. **Add a bypass**

   IP → **Hotspot** → **IP Bindings** tab → **+** → MAC Address: your PC's MAC → Type: **bypassed** → Comment: `Owner PC` → OK. Repeat for your own phone.

   ```
   /ip hotspot ip-binding add mac-address=AA:BB:CC:DD:EE:FF type=bypassed comment="Owner PC"
   ```

   (Replace AA:BB:CC:DD:EE:FF with the real MAC.)

✔ **You should see:** On your PC, any website opens without the login page.

Careful: Only add your own devices. Every bypassed device uses internet without paying and without speed limits.

PART 10

Hotspot settings

The settings the video recommends: one address per phone, and the login methods.

1 One address per phone

IP → **Hotspot** → **Servers** → double-click your server (**server1** or **hotspot1**) → Addresses Per MAC: `1` → OK.

```
/ip hotspot set [find] addresses-per-mac=1
```

2 Login methods

IP → **Hotspot** → **Server Profiles** tab → double-click **hsprof1** → **Login** tab → tick **HTTP CHAP**, **HTTP PAP**, **MAC Cookie**, **Cookie** (untick the others) → OK. (How long the MAC cookie lasts is set in the packages, Part 11.)

```
/ip hotspot profile set hsprof1 login-by=http-chap,http-pap,mac-cookie,cookie
```

Setting	Why
Addresses per MAC = 1	Each phone gets exactly one address.
HTTP PAP	Needed by your Starling login page (Part 12) and later the website's Connect button.
MAC Cookie	A phone that already logged in is let back in automatically after it reconnects, without typing again (until its time ends).

PART 11

Packages and prices (user profiles)

A “user profile” is a package: how long it lasts, how fast it is, how many phones can use it. Your six packages are ready in a file, including a small script that **starts the clock at the customer's first login and removes them automatically when the time is up**.

Profile	Length	Price	Phones	Speed (up/down)
`1hour`	1 hour	100 FCFA	1	2 / 5 Mbit/s
`12hours`	12 hours	150 FCFA	1	2 / 5 Mbit/s
`24hours`	24 hours	200 FCFA	1	2 / 5 Mbit/s
`1week`	7 days	1 350 FCFA	1	2 / 5 Mbit/s
`2weeks`	14 days	2 500 FCFA	1	2 / 5 Mbit/s
`1month`	30 days	4 500 FCFA	2	2 / 5 Mbit/s
`family`	never expires	free	2	5 / 10 Mbit/s

Tip: **family** is only for the family members who contributed to the Starlink (Part 13). Public customers can never get it unless you create a family login for them.

1 Upload the file

Files → drag `starling-profiles.rsc` from your PC into the Files window (or click **Upload** and choose it).

✔ **You should see:** The Files list shows `starling-profiles.rsc` at the top level (not inside a folder).

2 Import it

New Terminal → type:

```
/import file-name=starling-profiles.rsc
```

✔ **You should see:** Seven lines like “Profile 1hour (1h, 100 FCFA, 1 device(s)) ready” and “Profile family (free, never expires…) ready” and “Script file loaded and executed successfully”. **IP → Hotspot → User Profiles** lists the six profiles.

✖ **If not:** “no such file”: the file is not at the top level or the name is different; check **Files**. To try again, simply run the import again: it replaces the profiles.

Tip: Already imported an older version of this file? Upload the new one (replace) and run the import again. It updates the profiles in place and adds **family**; customers who are online keep working.

What each profile does (you do not need to change anything)

Setting	Value and meaning
Shared Users	**1**: one phone at a time per username (2 for 1month).
Rate Limit	**2M/5M**: 2 Mbit/s upload, 5 Mbit/s download per customer, so one person downloading a film does not slow everyone.
Idle Timeout	**none**: a customer is not logged out just for being idle.
Keepalive Timeout	**3h**: if a phone switches Wi-Fi off, it keeps its place for 3 hours, so the same username cannot be passed to a friend's phone.
Status Autorefresh	**1m**: the “time left” page refreshes every minute.
Scripts → On Login	At the first login: creates a timer (in **System → Scheduler**, named `exp-username`) and writes the start and end time in the user's Comment. When the timer ends: disconnects the phone and deletes the user.

Tip: To change a speed later: **IP → Hotspot → User Profiles** → double-click the profile → Rate Limit (e.g. `3M/8M`) → OK. To change a duration, edit the number after `interval=` in the profile's **Scripts → On Login** box (e.g. `1h`, `2d`, `7d`).

PART 12

Your Juthelink pages and automatic logins

Two things go on the router: the three Juthelink pages (what customers see) and one small script that collects the logins customers buy on the website, every 10 seconds. Together they use about 15 KB.

Tip: Where are the files? In `Juthelink-Router-Kit.zip`: the folder `hotspot` (`login.html`, `alogin.html`, `status.html`) and the file `juthelink-sync.rsc`. They are already made for your website **starling-website-lilac.vercel.app**. No editing needed.

Careful: Keep `juthelink-sync.rsc` private. It contains the router's secret key for the website. Do not send it on WhatsApp or post it anywhere. If it leaks, ask for a new key and a new file.

1 Unzip the kit on your PC

Right-click the zip → Extract. You get the PDF, `juthelink-sync.rsc`, `starling-profiles.rsc`, `ticket-maker.html` and the `hotspot` folder.

2 Upload the three pages into the hotspot folder

Winbox → **Files** → drag `login.html`, `alogin.html` and `status.html` from the kit's `hotspot` folder onto the **hotspot** folder (or open the folder and click **Upload**). Replace the old files when asked.

✔ **You should see:** `hotspot/login.html`, `hotspot/alogin.html`, `hotspot/status.html` with today's date, **inside** the hotspot folder.

✖ **If not:** They appear at the top level: drag them onto the folder name, or remove and upload again. Close and reopen Files if the list does not refresh.

3 Remove the logout page

Delete `hotspot/logout.html` (customers must not log out and pass their login to someone else). Leave every other file in the hotspot folder: the router uses some of them itself. To get MikroTik's original pages back at any time: **IP → Hotspot → Server Profiles → Reset HTML**.

4 Install the automatic-login script

Drag `juthelink-sync.rsc` into the Files window at the **top level** (not into the hotspot folder). Then open **New Terminal** and type:

```
/import file-name=juthelink-sync.rsc
```

✔ **You should see:** `Juthelink sync installed: the router now checks the website every 10 seconds`. In **System → Scheduler** there is **juthelink-sync** with interval **00:00:10**, and in **IP → Hotspot → Walled Garden IP List** there is **starling-website-lilac.vercel.app** (customers can open the website before they log in).

✖ **If not:** "failure" or nothing happens: the file must be at the top level in Files and named exactly `juthelink-sync.rsc`. Running the import again is safe.

5 Make sure both login methods are on

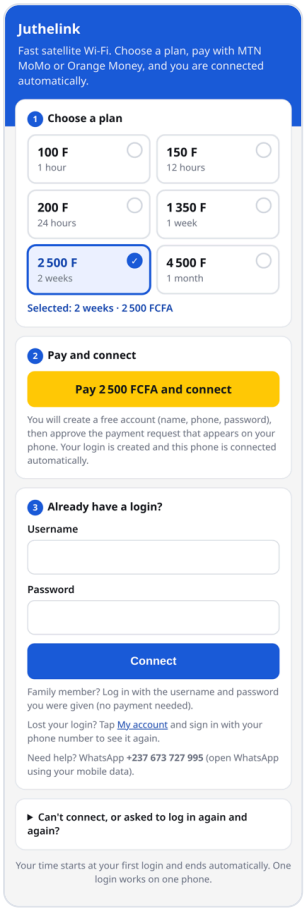
In the terminal:

```
/ip hotspot profile set [find] login-by=http-chap,http-pap,cookie,mac-cookie
```

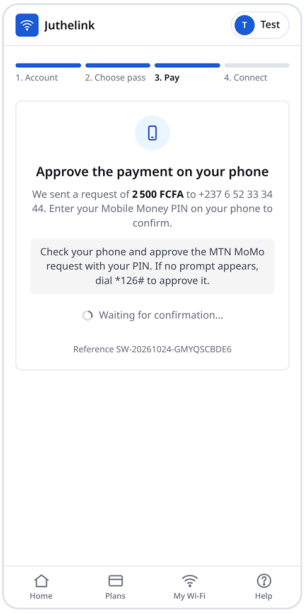
(The Juthelink page works with either; having both on avoids surprises.)

6 Check the router reaches the website

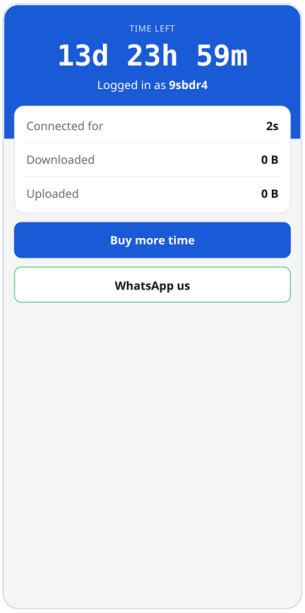
Wait 30 seconds, then **Log**. There must be **no** red line “juthelink-sync: website not reachable”. If there is, check the router has internet (`/ping google.com count=3`) and the clock is right (Part 3).



Login page (plan ticked)



Approve on the phone



Connected, time left

PART 13

How selling works (automatic, with Campay)

Customers pay with MTN MoMo or Orange Money through Campay. The website checks the payment with Campay itself, creates the login, and the router picks it up within about 10 seconds. The phone is then logged in automatically. You do nothing.

#	What the customer does / sees
1	Joins the Wi-Fi. The Juthelink page opens. Taps a plan: it gets a ✓ and “Selected: …”.
2	Taps **Pay … FCFA and connect**. The Juthelink website opens (allowed before login).
3	First time: creates a free account (name, phone, password). Next time: signs in.
4	Taps **Pay**. A MoMo / Orange Money prompt appears on their phone; they enter their PIN. (No prompt? Dial ***126#** for MTN or **#150*50#** for Orange to approve.)
5	“Payment confirmed”. A few seconds later the phone goes back to the Juthelink page, the username and password are **filled in automatically** and it connects by itself.
6	“You are connected”, then the **Time left** page counting down. When the time is up the router disconnects them and deletes the login.

Tip: A payment counts only when Campay confirms it to the website, for the exact amount. A customer cannot get internet by faking a page or a screenshot. A declined or cancelled payment gives nothing; the customer can try again.

Where do the logins come from?

The website makes a random username (6 letters/numbers, e.g. `k7m4qx`) and a 6-digit password for each purchase. The router adds it with the right package and puts the customer's phone number in the **comment**. At the first login the router adds the start and end time to that comment. You can see every login in **IP → Hotspot → Users**.

“I lost my login” / changed phone

On the Juthelink page: **Lost your login? → My account**, sign in with their phone number and password: the login is shown again with a **Connect** button. Forgot their password too? They tap “Forgot password” and use their Wi-Fi username or a code you give them from the admin site.

Tip: A customer who already logged in on a phone is usually let back in automatically (MAC cookie) until their time ends.

Where is the money?

In your Campay account (campay.net → your app **starling** → Money balance / Transactions). Every Juthelink payment has a reference like `SW-20261024-XXXXXXXX` that you also see in the admin site under Payments. Withdraw from Campay to your MoMo as usual.

Family members (free access)

For family members who contributed to buying the Starlink: open **ticket-maker.html** → Plan: **Family member (free)** → type their name (and phone) → **Make ticket** → paste the command into Winbox **New Terminal** → send them the message. Example:

```
/ip hotspot user add name=marie766 password=261043 profile=family comment="FAMILY: Marie Ngono, 677123456"
```

- Free, never expires, up to **2 of their own devices**, faster speed (5/10 Mbit/s).
- In **IP → Hotspot → Users** family logins have profile **family** and a comment starting with **FAMILY:**. **Find** → `FAMILY` lists them all.
- If a family login is shared with the public, change its password or click **✗ Disable**.

Cash customers and printed tickets

Someone pays you in cash? Use **ticket-maker.html** (works offline on your PC): choose the plan, **Make ticket**, paste the command into Winbox, give them the username and password. For tickets to sell later, leave the phone empty and choose e.g. 20 → **Print**. Each ticket's time starts at its first login.

PART 14

Tenda outdoor access point

The Tenda must only **broadcast the Wi-Fi** (“AP mode”). The MikroTik does everything else. If the Tenda acts as a router, all customers look like one phone and the hotspot cannot tell them apart.

14.1 Configure the Tenda on your desk first

1. **Wire it to your PC (not to the MikroTik yet)**
   Tenda cable → **PoE** port of the Tenda injector. Injector **LAN** port → your PC. Injector power → socket. Wait 1–2 minutes for the Tenda to start.

2. **Reset the Tenda**
   With power on, press and hold its reset button (or the button on the unit) for about 8 seconds, until the lights flash. Wait 1 minute.

3. **Give your PC an address in the Tenda's range**
   Look at the label on the Tenda for the default address (often `192.168.2.1`, user `admin`, password `admin`). On Windows: Settings → Network → Ethernet → IP assignment → Edit → **Manual**, IPv4 on, IP address `192.168.2.100`, Subnet mask `255.255.255.0`, Gateway empty → Save.

4. **Open the Tenda settings**
   Browser → `http://192.168.2.1` (or the address on the label) → log in. Set a new admin password if asked (write it in Part 0).

5. **Choose AP mode**
   Find **Working Mode / Operation Mode / Quick Setup** → choose **AP** (Access Point). **Not** Router, WISP, Client+AP or Repeater.

6. **Wi-Fi name and security**
   Wireless / Wi-Fi settings: SSID `Starling Wi-Fi` (exactly the same as the MikroTik), Security **None / Open**, Country **Cameroon** (if listed), Channel **1, 6 or 11** (choose a different one from the hAP lite if you keep both), Channel width **20 MHz**. If there is **Client Isolation / AP Isolation**, turn it **on**.

7. **LAN address and DHCP**
   LAN / Network settings: IP Address `10.5.50.2`, Subnet mask `255.255.255.0`, Gateway `10.5.50.1`, DNS `10.5.50.1`. **DHCP server: OFF** (if the option exists). Save. The Tenda reboots; your browser loses it. That is expected.

8. **Put your PC back to automatic**
   Windows: Ethernet → IP assignment → Edit → **Automatic (DHCP)** → Save.

14.2 Connect the Tenda to the MikroTik

MikroTik port 2 → cable → **Injector LAN** **Injector PoE** → long outdoor cable → **Tenda**

The injector also plugs into the power socket. Never plug the PoE port into the MikroTik or your PC.

1. **Plug in**
   Short cable from **MikroTik port 2** to the injector's **LAN** port. Keep the PoE port going to the Tenda.

2. **Check the Tenda is visible to the MikroTik**
   Winbox → **New Terminal**:

   ```
   /ping 10.5.50.2 count=4
   ```

✔ **You should see:** Four replies. From your PC (bypassed), `http://10.5.50.2` opens the Tenda settings.

✖ **If not:** No reply: check the cable is in port 2 and the injector's LAN port; redo 14.1 step 7 (address 10.5.50.2, gateway 10.5.50.1).

3. **Test through the Tenda**
   Stand near the Tenda (or turn off the hAP lite Wi-Fi for the test: **Wireless** → select wlan1 → **✗ Disable**). Phone → Starling Wi-Fi → open `http://neverssl.com`.

✔ **You should see:** The blue Starling login page. In Winbox, **IP → DHCP Server → Leases** shows your phone's own MAC address and a 10.5.50.x address.

✖ **If not:** The phone gets a 192.168.x.x address, or several phones show the same MAC in **IP → Hotspot → Hosts**: the Tenda is still in router mode or its DHCP is on. Redo 14.1 steps 5 and 7.

14.3 Mounting outdoors

- Mount it 3–6 m high on a pole or wall, facing where customers sit, antennas pointing up.
- Leave a “drip loop” (a U-shaped bend) in the cable below the Tenda so rain runs off, and close the cable cover tightly.
- Keep the injector indoors and dry.
- If you keep the hAP lite's own Wi-Fi on (indoors), use a different channel from the Tenda (e.g. 1 and 11). Same name is fine: phones pick the strongest.

PART 15

Final test checklist

Tick every line. Use a phone that is not bypassed (Part 9), with private Wi-Fi address turned off.

✓	Test	Expected
☐	Winbox → New Terminal → `/ping google.com count=4`	4 replies
☐	Phone joins **Starling Wi-Fi** near the Tenda	Connected, no password asked
☐	Open `http://neverssl.com`	Blue Juthelink login page
☐	Log in with a wrong password	Red message “invalid username or password”
☐	Log in with `test001` / `123456` (1hour)	“You are connected”, then Time left page
☐	Open YouTube / WhatsApp	Works
☐	A second phone tries the same username at the same time	Refused (“no more sessions are allowed”)
☐	Turn Wi-Fi off and on again on the first phone	Back online without typing (MAC cookie)
☐	Winbox **IP → Hotspot → Users**	test001 Comment shows Started … | ends …
☐	Wait 1 hour (or test with a `10m` profile)	Phone disconnected, login page appears again, test001 deleted
☐	Buy the cheapest plan with your own MoMo number on a phone that is not bypassed (Part 13)	Prompt on the phone → “Payment confirmed” → the page fills the login and connects by itself → Time left counting down
☐	Unplug the router power, plug back in	After 1 minute everything works again

Tip: Quick expiry test without waiting an hour: **IP → Hotspot → User Profiles → +** → Name `test10min`, Shared Users 1 → **Scripts** tab → paste the On Login script from `on-login-script.txt` and change `DURATION` to `10m` → OK. Create a user with this profile and log in. Delete the profile after the test.

Save your work (backup)

1. **Make a backup on the router and copy it to your PC**

   Files → **Backup** → Name `starling-working` → Backup. Then drag `starling-working.backup` from the Files window to your PC. Also in the terminal:

   ```
   /export file=starling-working
   ```

   and copy `starling-working.rsc` to your PC too (it is a readable text copy of all settings).

Tip: To go back to this working state one day: **Files** → select `starling-working.backup` → **Restore**.

PART 16

Daily use

I want to…	Where in Winbox
Sell a pass	Nothing to do: customers pay with Campay on the login page and are connected automatically (Part 13). Cash customer: ticket-maker.html.
See payments and money	Campay account → app **starling** → Transactions; the admin site → Payments.
Give a family member free access	ticket-maker.html → **Family member (free)** → name → paste the command (Part 13).
See all family logins	**IP → Hotspot → Users → Find** → `FAMILY`.
Find a customer's login (lost login)	The customer: Juthelink page → **My account**. You: **IP → Hotspot → Users → Find** → their phone number.
Check automatic logins are working	**System → Scheduler** → **juthelink-sync** Run Count goes up every 10 s; **Log** has no “website not reachable”.
See who is online now	**IP → Hotspot → Active** (user, address, uptime, data).
See when a customer's time ends	**IP → Hotspot → Users** → Comment column (“Started … | ends …”), or **System → Scheduler** → `exp-username` → Next Run.
Disconnect a phone (e.g. customer changed phone)	**IP → Hotspot → Active** → select the line → **−** (Remove). The customer can log in again on the new phone; their time keeps running.
Stop a customer completely	**IP → Hotspot → Users** → select → **✗ Disable**, then remove them from Active.
Give someone free internet (staff)	**IP → Hotspot → IP Bindings → +**, their MAC, type bypassed.
Change prices on the login page	Edit login.html on your PC and upload it again into the hotspot folder (Part 12).
Check the router is healthy	Bottom bar of Winbox: CPU % and memory. CPU often above 70 % when busy means it is time for a bigger router.

Tell every customer

- **Turn off private / random Wi-Fi address** for Starling Wi-Fi. Android: Settings → Wi-Fi → gear → Privacy → **Use device MAC**. iPhone: Settings → Wi-Fi → ⓘ → turn off **Private Wi-Fi Address**. Otherwise the Wi-Fi keeps asking them to log in.
- Type the username and password exactly, with no spaces.
- One ticket works on one phone, and the time starts at the first login.

PART 17

Problems and fixes

Problem	Fix
Winbox cannot find the router	Use the MAC address. Replug the yellow cable. Try another port. Last resort: hardware reset (unplug power, hold reset, plug power, keep holding about 5 s until the light flashes, release), then start again at Part 2.
DHCP Client on ether1 not “bound”	Blue cable in port 1? Starlink app Online? Restart the Starlink router.
Customers get internet without logging in	FastTrack rule active (Part 8 step 2), or the device is bypassed (Part 9), or the hotspot server hotspot1 is disabled.
Grey MikroTik page instead of the blue one	Files not inside the hotspot folder or wrong name (Part 12).
No login page pops up	Open `http://neverssl.com` or `http://login.starling`. HTTPS sites (Google, Facebook) cannot show the login page; that is normal.
“invalid username or password”	Extra space or capital letter. Check the user in IP → Hotspot → Users. The user may have expired (deleted).
“no more sessions are allowed for user”	Another phone is using this ticket (or the same phone with a new random MAC). Remove it from Active, and ask the customer to turn off private Wi-Fi address.
Customer asked to log in again and again	Private / random Wi-Fi address is on. See Part 16.
Phones get 192.168.x.x addresses	Tenda DHCP is on or Tenda is in router mode (Part 14.1 steps 5 and 7).
All phones show the same MAC in IP → Hotspot → Hosts	Tenda is in router mode. Put it in AP mode.
Time ended too early or too late	Clock wrong: System → Clock and System → SNTP Client (Part 3 step 3). Check internet works.
“Pay … and connect” opens a blank / error page	The website is not in the walled garden: import `juthelink-sync.rsc` again (Part 12 step 4). Check the router has internet and DNS (`/ping starling-website-lilac.vercel.app`).
Paid, “Payment confirmed”, but not connected after 1 minute	Log shows “website not reachable”: router internet or clock problem (Part 3, Part 4). Or the scheduler **juthelink-sync** is disabled. The customer can tap **My account** → **Connect** once it is fixed; the login is kept.
Log: “juthelink-sync: could not add … (profile …)”	The package profiles are missing: import `starling-profiles.rsc` again (Part 11).
“We could not start the payment”	The number is not a MoMo / Orange Money number, or Campay is down. Ask the customer to try again or use another number.
Profiles import fails	The file must be at the top level in Files. Run the import again.
Everything is slow	Check Starlink speed in the Starlink app. Lower the per-customer speed in the profiles. Check router CPU.

APPENDIX A

All commands in one place

Exactly what the clicks above do, for the terminal. Run them in order, one block at a time, after Part 2 (start clean). Replace the values in CAPITALS first.

```
# Part 3 - password, name, clock
/password old-password="" new-password="YOUR-PASSWORD" confirm-new-password="YOUR-PASSWORD"
/system identity set name=starling-hotspot
/system clock set time-zone-name=Africa/Douala
/system ntp client set enabled=yes primary-ntp=162.159.200.1 secondary-ntp=162.159.200.123

# Part 4 - internet from Starlink on port 1
/ip dhcp-client add interface=ether1 use-peer-dns=no add-default-route=yes disabled=no
comment="Internet from Starlink"
/ip dns set servers=1.1.1.1,8.8.8.8 allow-remote-requests=yes
/ping google.com count=4

# Part 6 - customer bridge
/interface bridge add name=bridge-hotspot comment="Customers"
/interface bridge port add bridge=bridge-hotspot interface=ether2
/interface bridge port add bridge=bridge-hotspot interface=ether3
/interface bridge port add bridge=bridge-hotspot interface=ether4
/interface bridge port add bridge=bridge-hotspot interface=wlan1
/ip address add address=10.5.50.1/24 interface=bridge-hotspot comment="Customer gateway"

# Part 7 - the hAP lite's own open Wi-Fi
/interface wireless security-profiles set [find default=yes] mode=none
/interface wireless set wlan1 mode=ap-bridge band=2ghz-b/g/n channel-width=20mhz frequency=auto
ssid="Starling Wi-Fi" security-profile=default wireless-protocol=802.11 disabled=no

# Part 8 - hotspot (what the Hotspot Setup wizard creates)
/ip pool add name=hs-pool ranges=10.5.50.10-10.5.50.250
/ip dhcp-server add name=dhcp-hotspot interface=bridge-hotspot address-pool=hs-pool lease-
time=1h disabled=no
/ip dhcp-server network add address=10.5.50.0/24 gateway=10.5.50.1 dns-server=10.5.50.1
/ip hotspot profile add name=hsprof1 hotspot-address=10.5.50.1 dns-name=login.starling html-
directory=hotspot smtp-server=0.0.0.0
/ip hotspot add name=hotspot1 interface=bridge-hotspot address-pool=hs-pool profile=hsprof1
disabled=no
/ip hotspot user add name=admin password="YOUR-TEST-PASSWORD"
/ip firewall nat add chain=srcnat src-address=10.5.50.0/24 action=masquerade
comment="masquerade hotspot network"
/ip firewall filter disable [find action=fasttrack-connection]

# Part 9 - your own devices skip the login
/ip hotspot ip-binding add mac-address=YOUR:PC:MAC:AA:BB:CC type=bypassed comment="Owner PC"

# Part 10 - hotspot settings
/ip hotspot set [find] addresses-per-mac=1
/ip hotspot profile set hsprof1 login-by=http-chap,http-pap,mac-cookie,cookie

# Part 11 - packages (upload starling-profiles.rsc to Files first)
/import file-name=starling-profiles.rsc

# Part 13 - a customer
/ip hotspot user add name=frank482 password=739215 profile=24hours

# Backup
/export file=starling-working
```

Tip: Every command on this page was tested on RouterOS 6.48 before this guide was written. When you create the hotspot with the terminal (Part 8) instead of the wizard, the `hotspot` folder with MikroTik's default pages is created automatically, exactly as with the wizard.

Phase 2 (later): the Starling website and payments connect to this same router through a secure tunnel. It uses its own profiles and users and never touches the ones you create here. See `docs/network-integration-guide.md` Part 8.